

CONFIGURACION DE REPLICACION EN MONGO

1. Importación de la Clave Pública de MongoDB

wget -qO - https://www.mongodb.org/static/pgp/server-6.0.asc | sudo apt-key add

```
dany8mt@master:~$ wget -qO - https://www.mongodb.org/static/pgp/server-6.0.asc | sudo apt-key add
Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (see apt-key(8)).
OK
```

2. Configuración del Repositorio

echo "deb [arch=amd64,arm64] https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/6.0 multiverse" | sudo tee /etc/apt/sources.list.d/mongodb-org-6.0.list

```
dany8mt@master:~$ echo "deb [ arch=amd64,arm64 ] https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/6.0 multiverse" | sudo tee /etc/apt/sources.list.d/mongodb-org-6.0.list
deb [ arch=amd64,arm64 ] https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/6.0 multiverse
```

3. Actualización e Instalación

sudo apt-get update

sudo apt-get install -y mongodb-org

```
dany8mt@master:~$ sudo apt-get install -y mongodb-org
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  mongodb-database-tools mongodb-mongosh mongodb-org-database mongodb-org-database-tools-extra
  mongodb-org-mongos mongodb-org-server mongodb-org-shell mongodb-org-tools
The following NEW packages will be installed:
  mongodb-database-tools mongodb-mongosh mongodb-org mongodb-org-database mongodb-org-database-tools-extra
  mongodb-org-mongos mongodb-org-server mongodb-org-shell mongodb-org-tools
0 upgraded, 9 newly installed, 0 to remove and 12 not upgraded.
Need to get 168 MB of archives.
```

4. Inicialización del Servicio

sudo systemctl start mongod

sudo systemctl enable mongod

```
dany8mt@master:~$ sudo systemctl status mongod
● mongod.service - MongoDB Database Server
   Loaded: loaded (/lib/systemd/system/mongod.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2025-06-11 17:03:10 UTC; 6s ago
     Docs: https://docs.mongodb.org/manual
   Main PID: 4288 (mongod)
    Memory: 177.6M
      CPU: 1.778s
    CGroup: /system.slice/mongod.service
            └─4288 /usr/bin/mongod --config /etc/mongod.conf

Jun 11 17:03:10 master systemd[1]: Started MongoDB Database Server.
Jun 11 17:03:10 master mongod[4288]: {"t":{"$date":"2025-06-11T17:03:10.724Z"},"s":"I", "c":"CON
...skipping...
```

Configuración de Replicación

1. Modificación del Archivo de Configuración

Has añadido estas líneas al archivo de configuración de MongoDB (ubicado en /etc/mongod.conf):

YAML

replication:

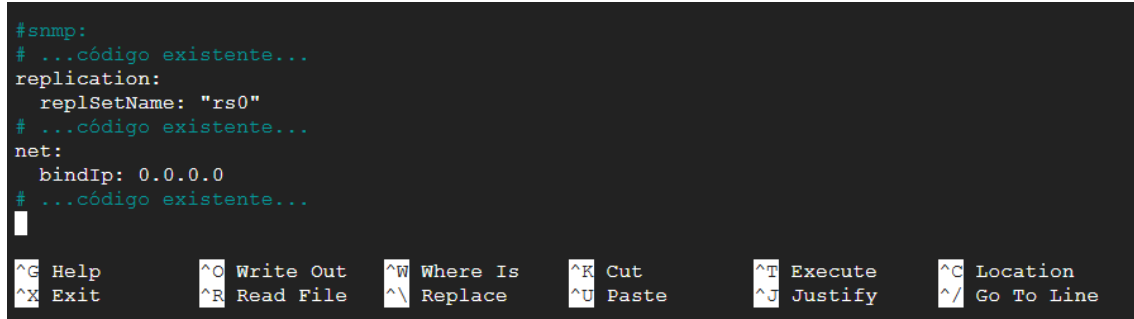
replSetName: "rs0"

net:

bindIp: 0.0.0.0

```
#snmp:
# ...código existente...
replication:
  replSetName: "rs0"
# ...código existente...
net:
  bindIp: 0.0.0.0
# ...código existente...

```



2. Reinicio del Servicio

sudo systemctl restart mongod

Captura de pantalla recomendada #6: Resultado de systemctl status mongod después del reinicio.

3. Inicialización del Replica Set

```
rs.initiate({
  _id: "rs0",
  members: [
    { _id: 0, host: "34.172.190.71:27017" },
    { _id: 1, host: "104.154.217.212:27017" },
    { _id: 2, host: "34.9.104.16:27017" }
  ]
})
```

```

test>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
test> rs.initiate({
...   _id: "rs0",
...   members: [
...     { _id: 0, host: " 34.172.190.71:27017" },
...     { _id: 1, host: " 104.154.217.212:27017" },
...     { _id: 2, host: " 34.9.104.16:27017" }
...   ]
... })
...
{
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1749661743, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1749661743, i: 1 })
}
rs0 [direct: secondary] test>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)

```

4. Verificación del Estado del Replica Set

rs.status()

```

rs0 [direct: secondary] test> rs.status()
{
  set: 'rs0',
  date: ISODate('2025-06-11T17:09:28.690Z'),
  myState: 1,
  term: Long('1'),
  syncSourceHost: '',
  syncSourceId: -1,
  heartbeatIntervalMillis: Long('2000'),
  majorityVoteCount: 2,
  writeMajorityCount: 2,
  votingMembersCount: 3,
  writableVotingMembersCount: 3,
  optimes: {
    lastCommittedOpTime: { ts: Timestamp({ t: 1749661756, i: 6 }), t: Long('1') },
    lastCommittedWallTime: ISODate('2025-06-11T17:09:16.334Z'),
    readConcernMajorityOpTime: { ts: Timestamp({ t: 1749661756, i: 6 }), t: Long('1') },
    appliedOpTime: { ts: Timestamp({ t: 1749661756, i: 6 }), t: Long('1') },
    durableOpTime: { ts: Timestamp({ t: 1749661756, i: 6 }), t: Long('1') },
    lastAppliedWallTime: ISODate('2025-06-11T17:09:16.334Z'),
    lastDurableWallTime: ISODate('2025-06-11T17:09:16.334Z')
  },
  lastStableRecoveryTimestamp: Timestamp({ t: 1749661743, i: 1 }),
  electionCandidateMetrics: {
    lastElectionReason: 'electionTimeout',
    lastElectionDate: ISODate('2025-06-11T17:09:15.022Z'),
    electionTerm: Long('1'),
    lastCommittedOpTimeAtElection: { ts: Timestamp({ t: 1749661743, i: 1 }), t: Long('-1') },
    lastSeenOpTimeAtElection: { ts: Timestamp({ t: 1749661743, i: 1 }), t: Long('-1') },
    numVotesNeeded: 2,

```

Análisis de la Configuración

La configuración implementada tiene las siguientes características:

1. **Versión de MongoDB:** 6.0 (última versión estable)
2. **Sistema Operativo:** Ubuntu 22.04 (Jammy)

3. **Nombre del Replica Set:** rs0
4. **Número de Nodos:** 3 (configuración recomendada para alta disponibilidad)
5. **Configuración de Red:** Escucha en todas las interfaces (0.0.0.0)

Estructura del Replica Set

Nodo ID	Dirección IP	Puerto	Rol Inicial Esperado
0	34.172.190.71	27017	PRIMARY (probable)
1	104.154.217.212	27017	SECONDARY
2	34.9.104.16	27017	SECONDARY

```
rs0 [direct: primary] test> rs.isMaster()
{
  topologyVersion: {
    processId: ObjectId('6849b6ce7d1186a9afcd68db'),
    counter: Long('6')
  },
  hosts: [
    '34.172.190.71:27017',
    '104.154.217.212:27017',
    '34.9.104.16:27017'
  ],
  setName: 'rs0',
  setVersion: 1,
  ismaster: true,
  secondary: false,
  primary: '34.172.190.71:27017',
  me: '34.172.190.71:27017',
  electionId: ObjectId('7fffffff0000000000000001'),
  lastWrite: {
    opTime: { ts: Timestamp({ t: 1749662175, i: 1 }), t: Long('1') },
    lastWriteDate: ISODate('2025-06-11T17:16:15.000Z'),
    majorityOpTime: { ts: Timestamp({ t: 1749662175, i: 1 }), t: Long('1') },
    majorityWriteDate: ISODate('2025-06-11T17:16:15.000Z')
  },
  maxBsonObjectSize: 16777216,
  maxMessageSizeBytes: 48000000,
  maxWriteBatchSize: 100000,
  localTime: ISODate('2025-06-11T17:16:22.861Z'),
  logicalSessionTimeoutMinutes: 30,
  connectionId: 2,
  minWireVersion: 0,
  maxWireVersion: 17,
  readOnly: false,
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1749662175, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
```

Recomendaciones de Seguridad

La configuración actual permite conexiones desde cualquier IP (bindIp: 0.0.0.0). Para entornos de producción, se recomienda:

1. Configurar un firewall para limitar el acceso solo a IPs confiables
2. Habilitar la autenticación interna de MongoDB
3. Configurar TLS/SSL para cifrar las comunicaciones entre nodos
4. Crear usuarios con privilegios específicos